

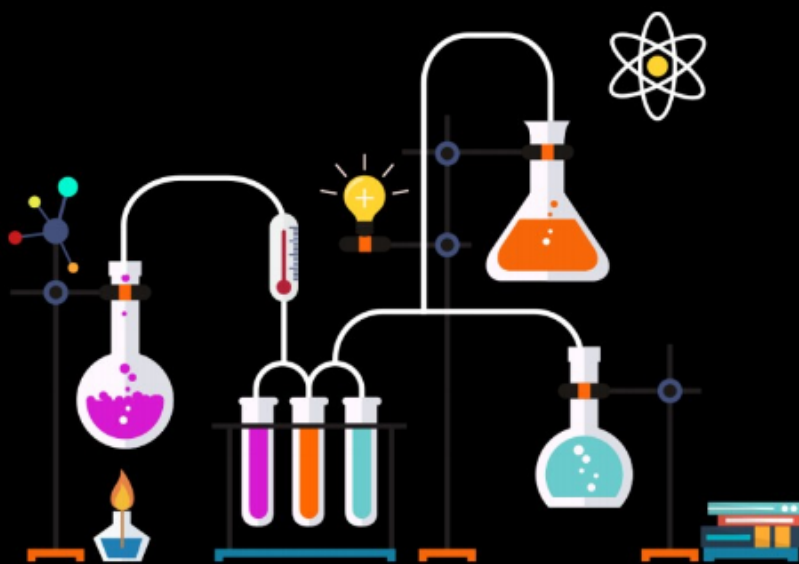
ALORON

জৈব রসায়ন

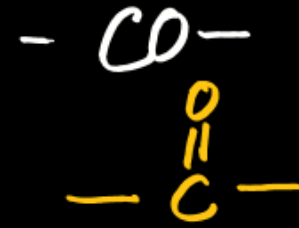
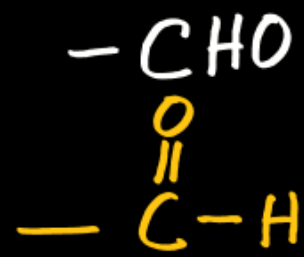
ORGANIC ZERO TO PRO

Lecture-19 (2nd Part)

Mottasin Pahlovi



অ্যালডিহাইড ও কিটোন



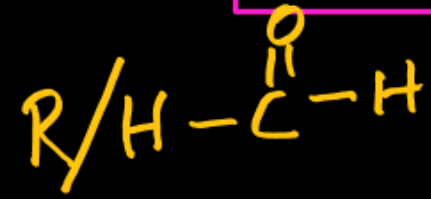
অ্যালডিহাইড ও কিটোন পরস্পরের

ব্যবহারী মূলক অণু

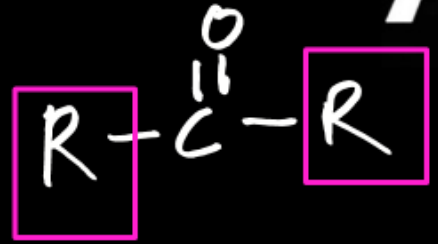
অ্যালডিহাইড

কিটোন



অ্যালডিহাইড
(al)

কিটোন



ফর্মালডিহাইড

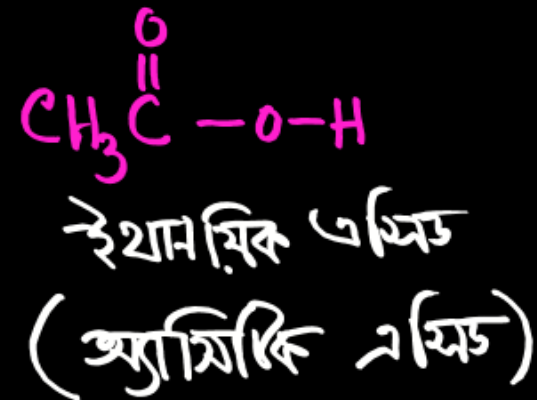
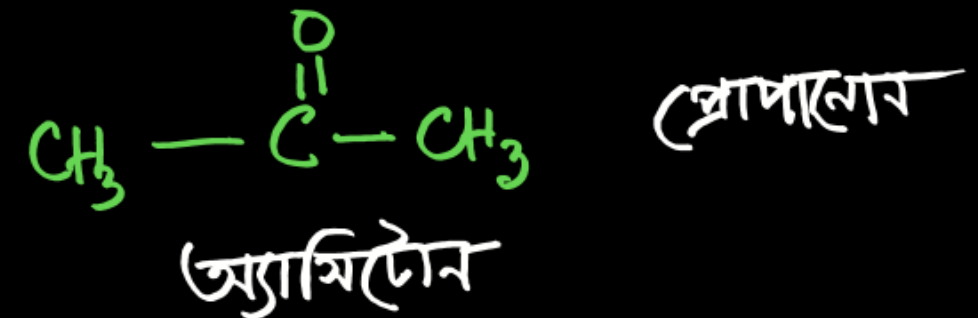


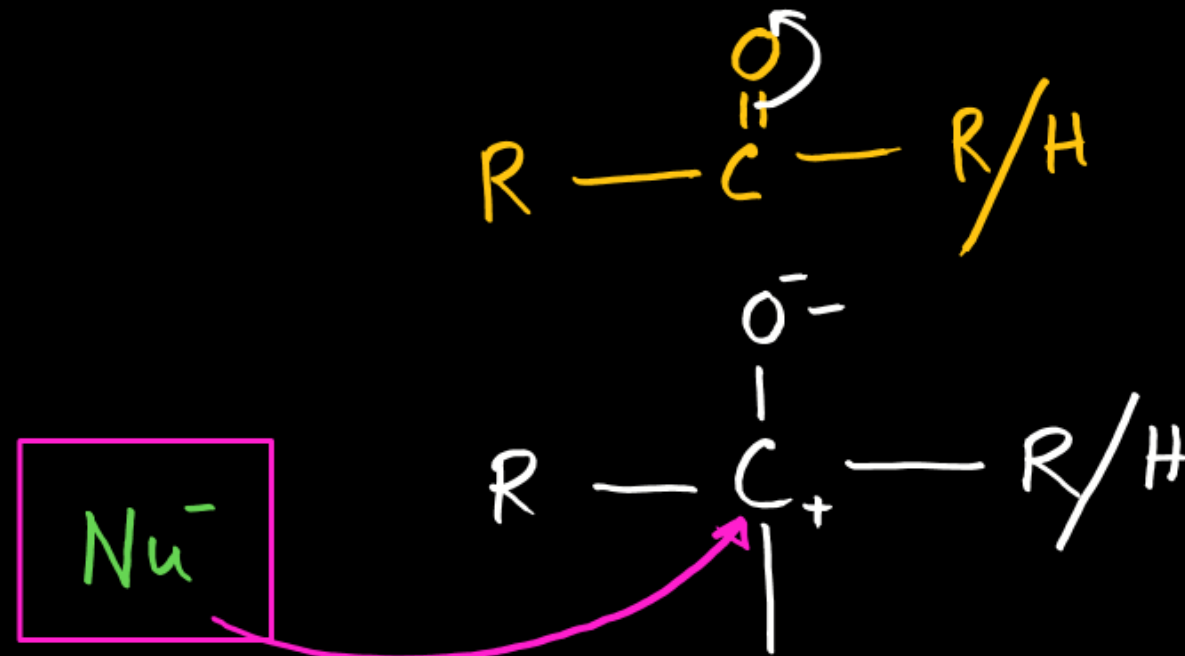
অ্যাসিটালডিহাইড



প্রোপ্যনালডিহাইড

অ্যামিটোনডিহাইড



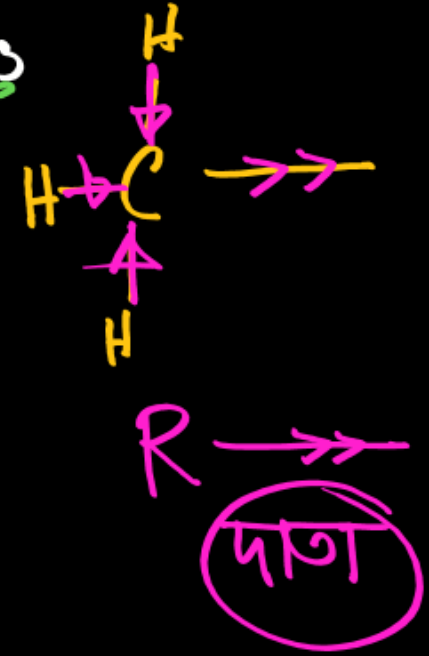
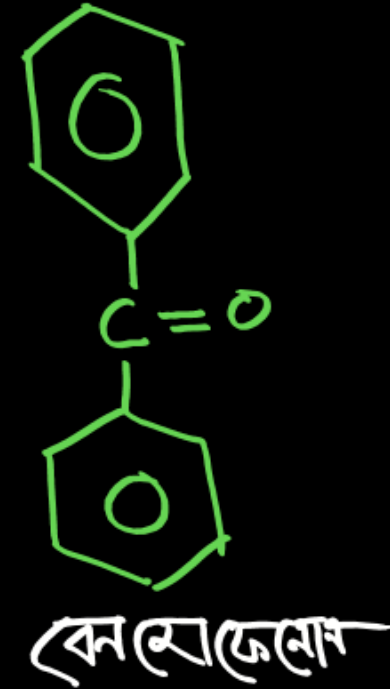
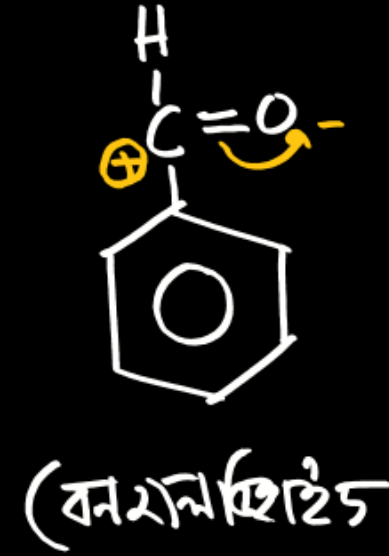
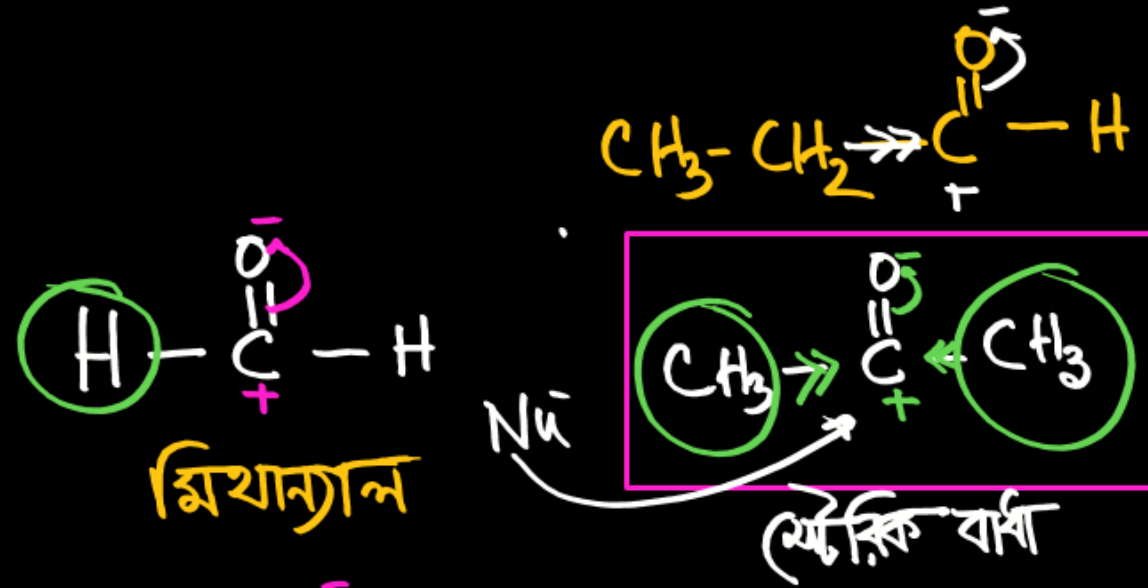
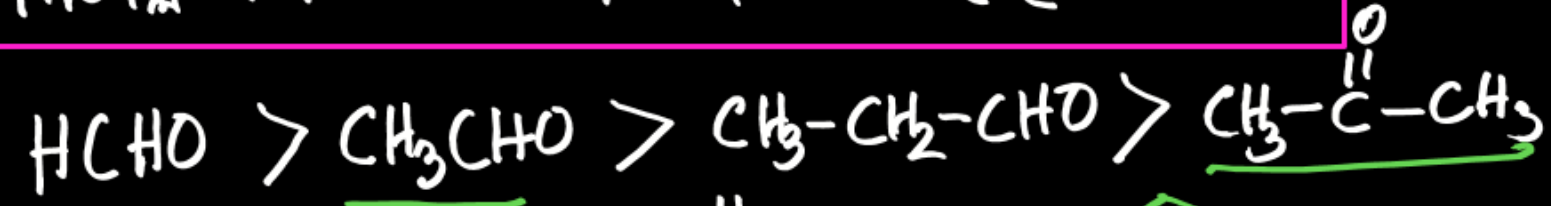


অ্যানিডাইড ও কিটোন
 নিউক্লিওফিলিক (কেন্দ্রাকর্ষী)
 ম্যথোজন বিক্রিয়া দেয়
 (যত)

MCA

R-
অ্যালকাইন যুক্তক
(e দাতা)

নিউক্লিওফিলিক বিক্রিয়া দ্বারা প্রভাবিত



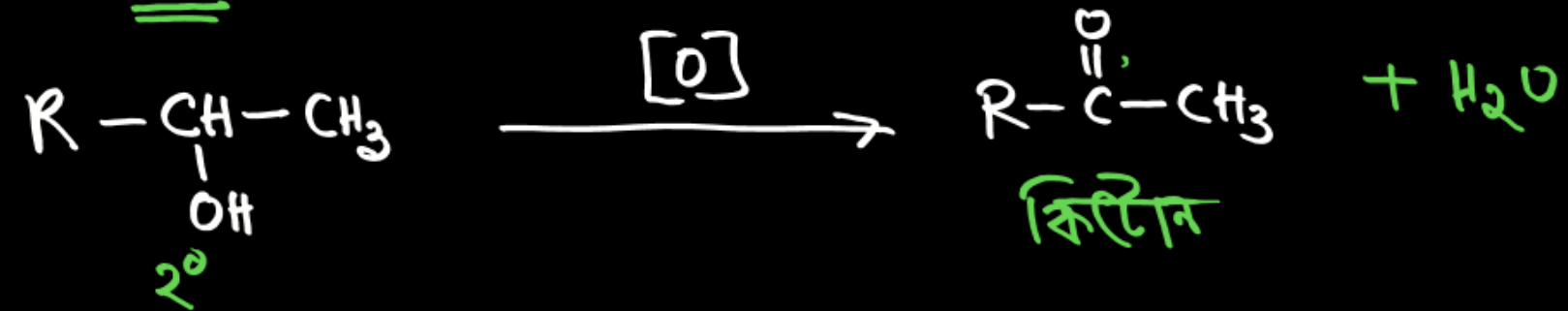
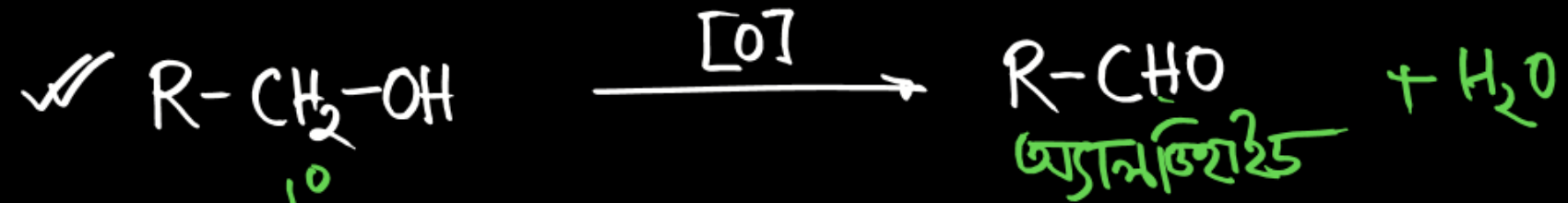
অ্যান্ডিহাইড ও কিটোনের প্রভৃতি

- ১/ অ্যালকোহল থেকে
- ২/ ডাই হ্যালাইড থেকে
- ৩/ অ্যালকিন থেকে
- ৪/ এমিড হ্যালাইড থেকে
- ৫/ এমিডের ক্যানমিয়ার্স নকল থেকে
- ৬/ অ্যাক্সাইন থেকে

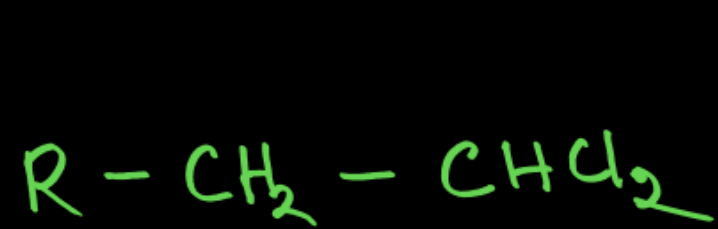
① অ্যালকোহল থেকে

জারক : $K_2Cr_2O_7 + H_2SO_4$ (লঘু)

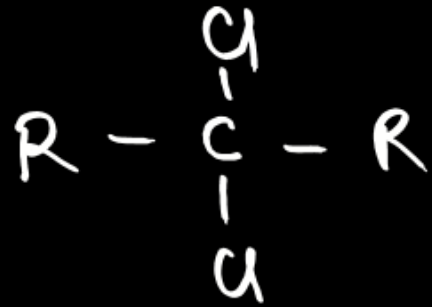
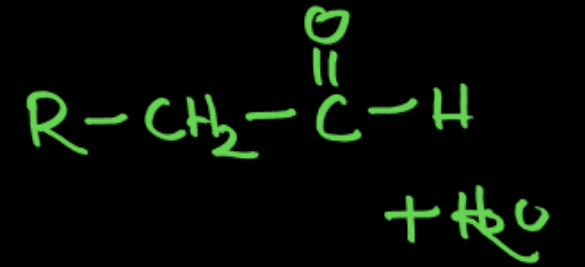
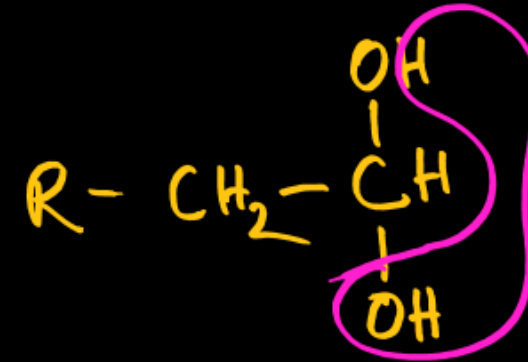
$KMnO_4 + H_2SO_4$ (লঘু)



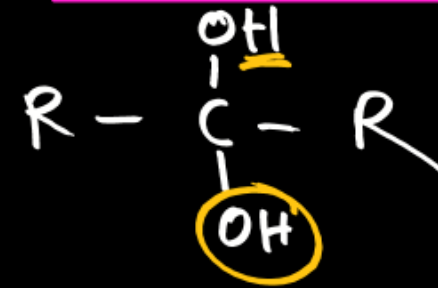
② এই শ্রানাইড ২২ক:



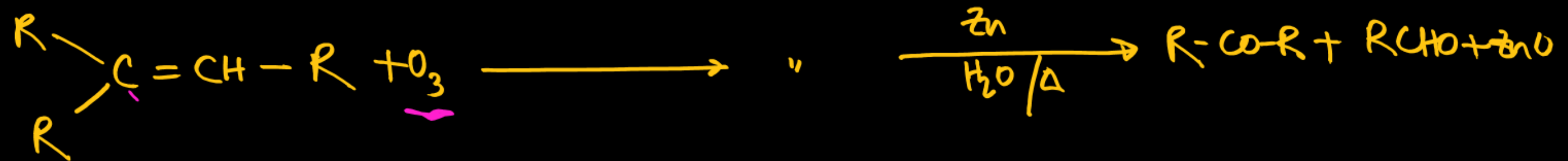
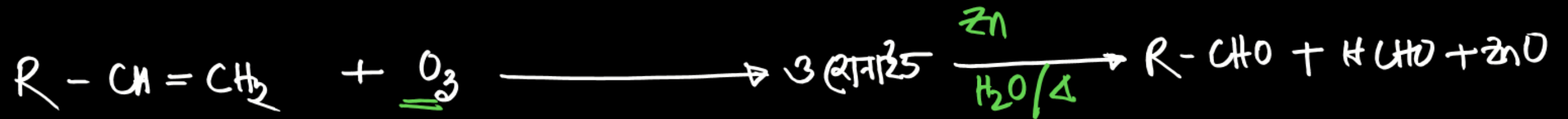
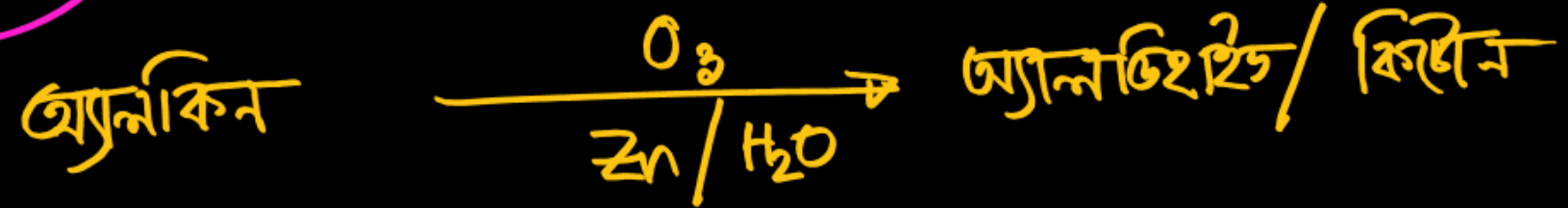
$\xrightarrow{NaOH(জলীয়)}$



$\xrightarrow{KOH(aq)}$

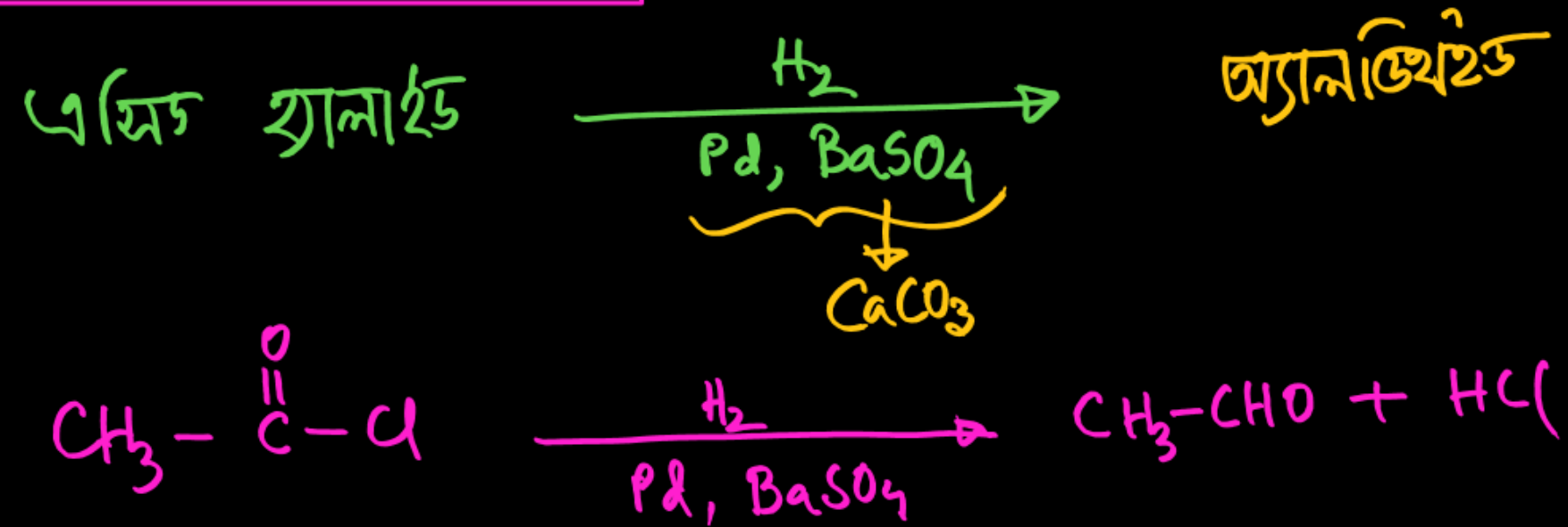


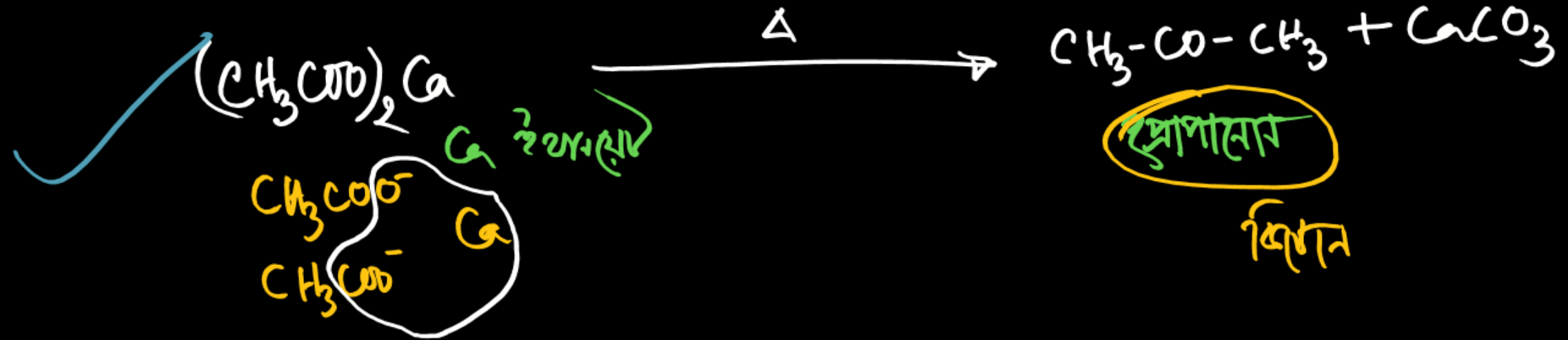
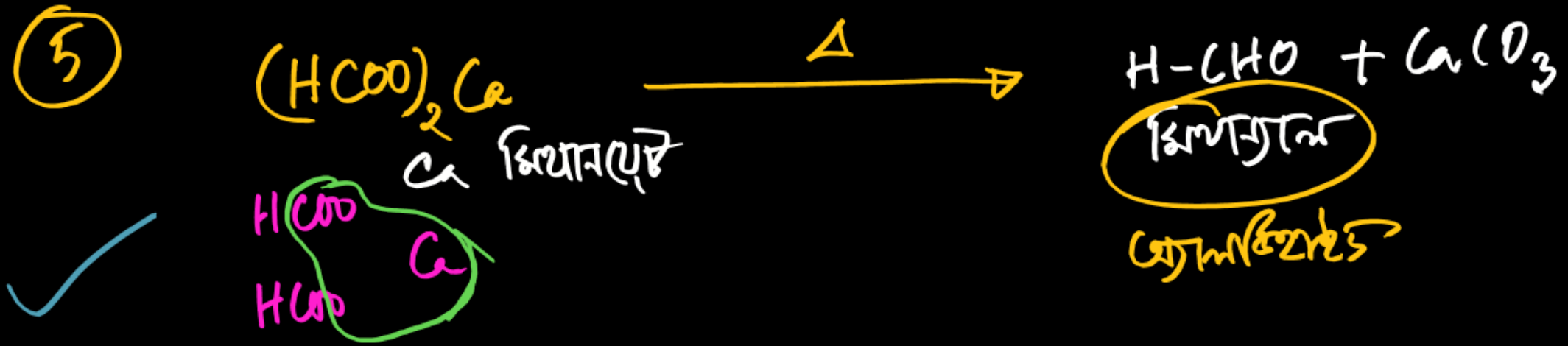
③ অ্যালকিন থেকে:

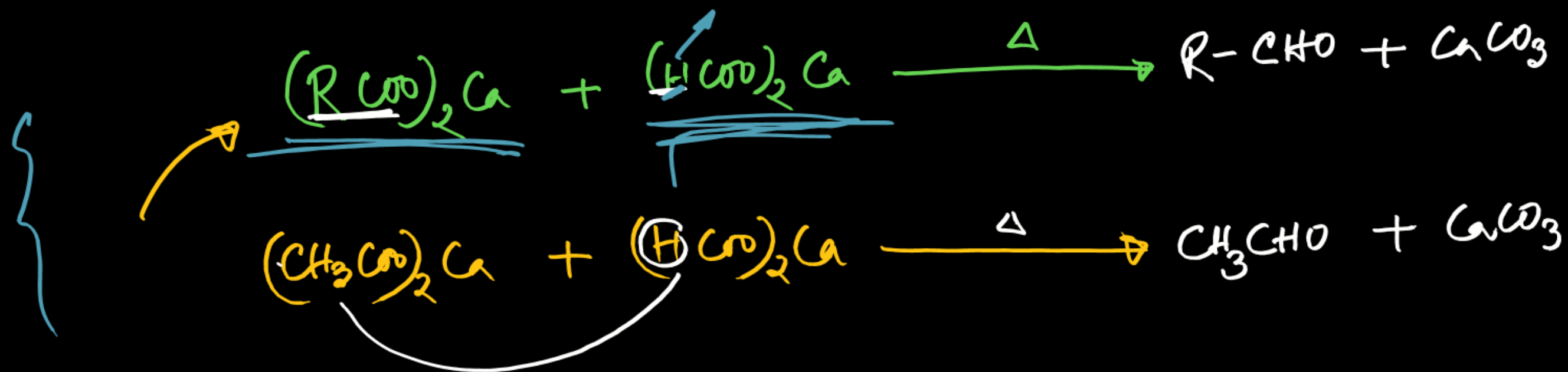


④ এসিড শানাইড থেকে:

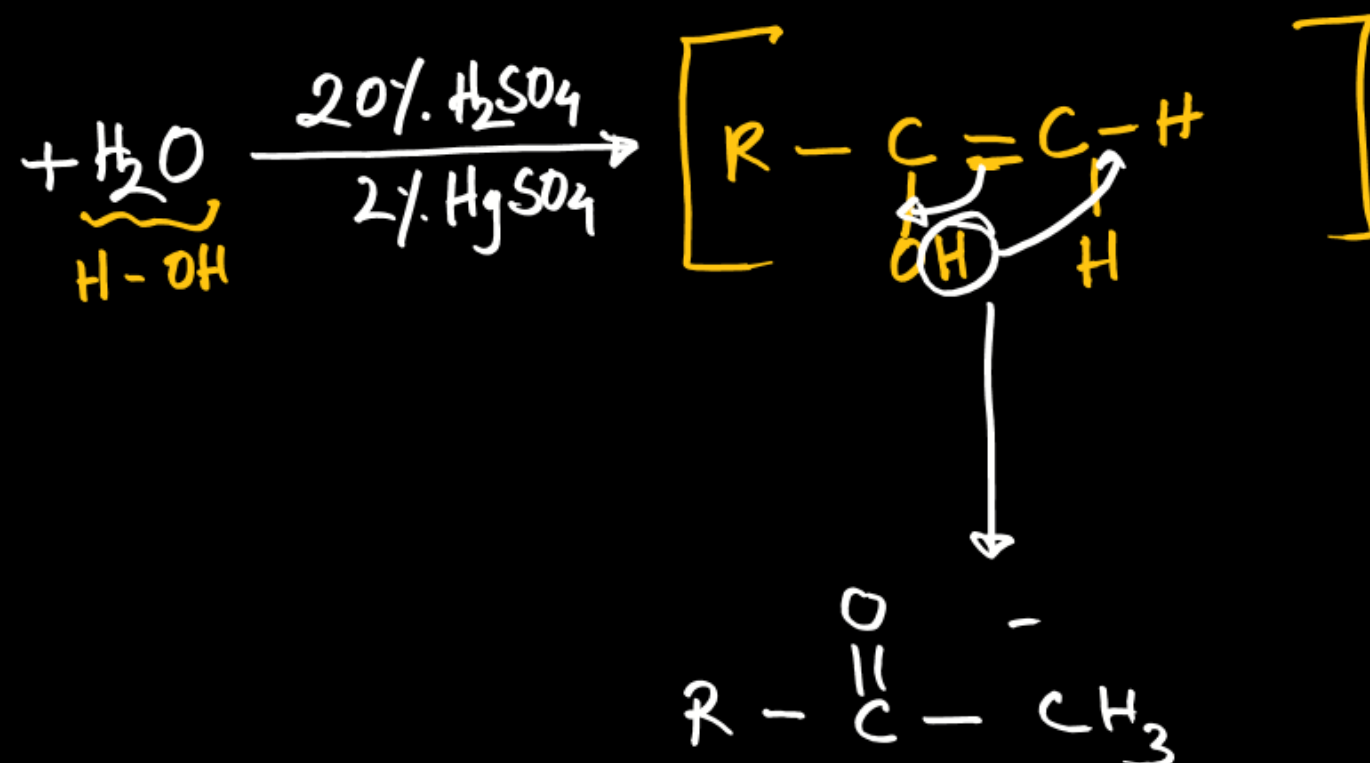
বাহ্যিক
বিভাগ







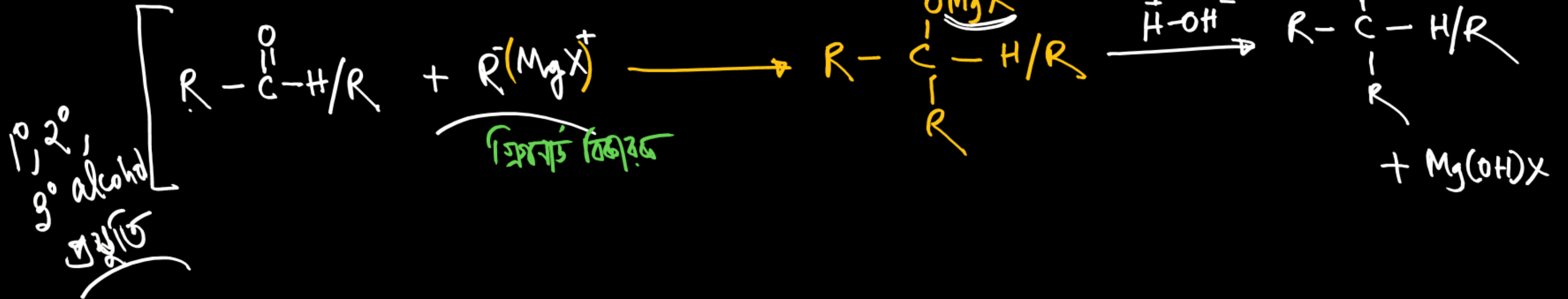
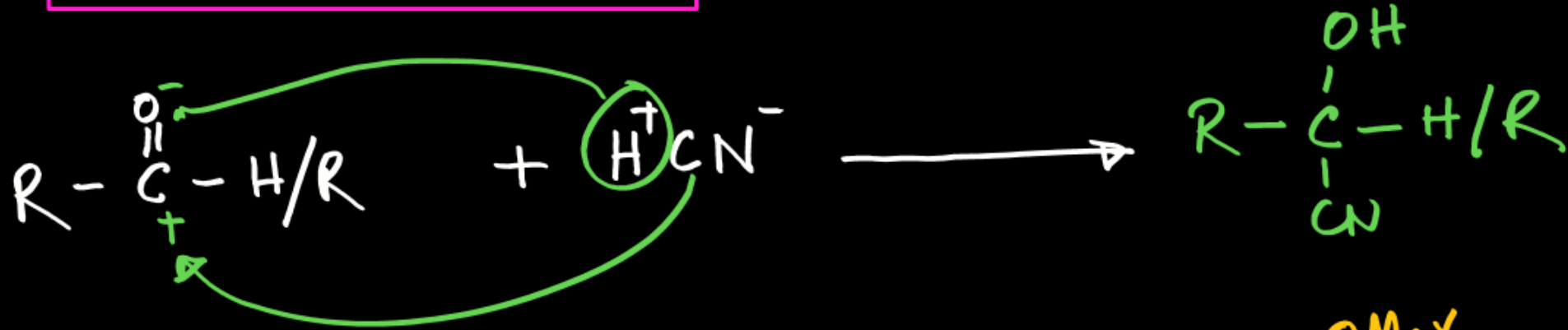
⑥ অ্যান্‌সাইন থেকে:



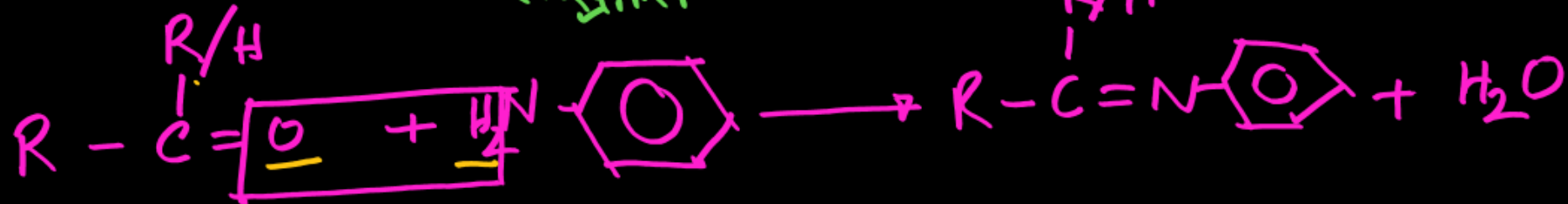
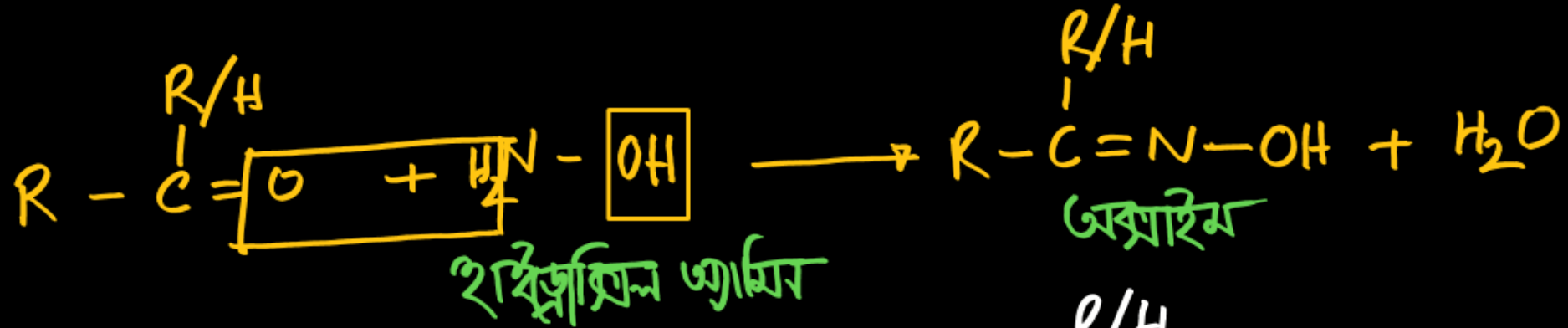
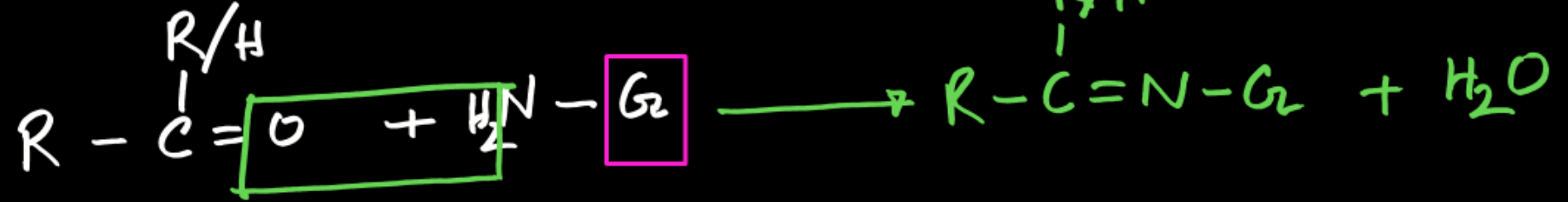
অ্যালডিহাইড ও কিটনের বিক্রিয়া

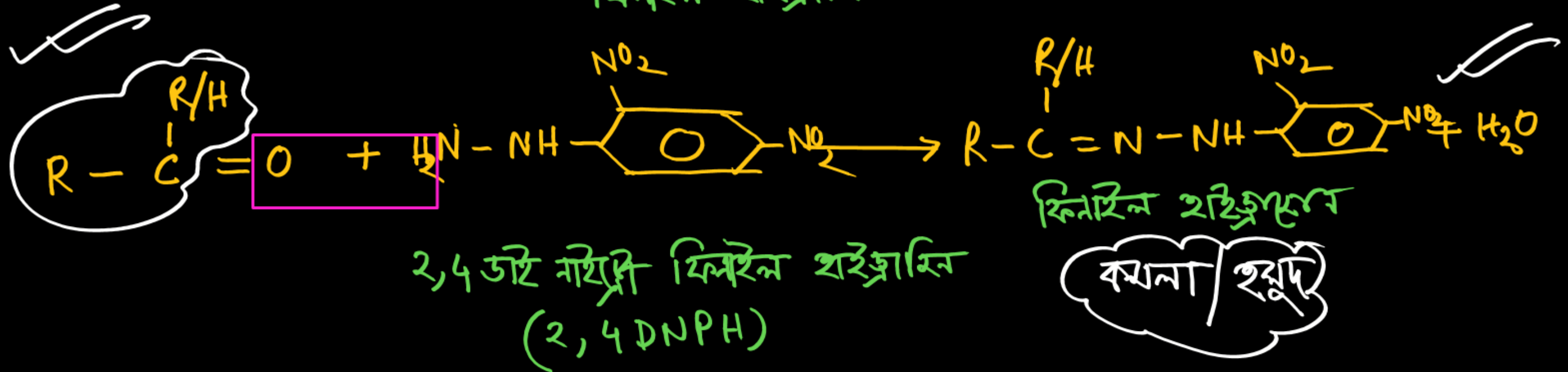
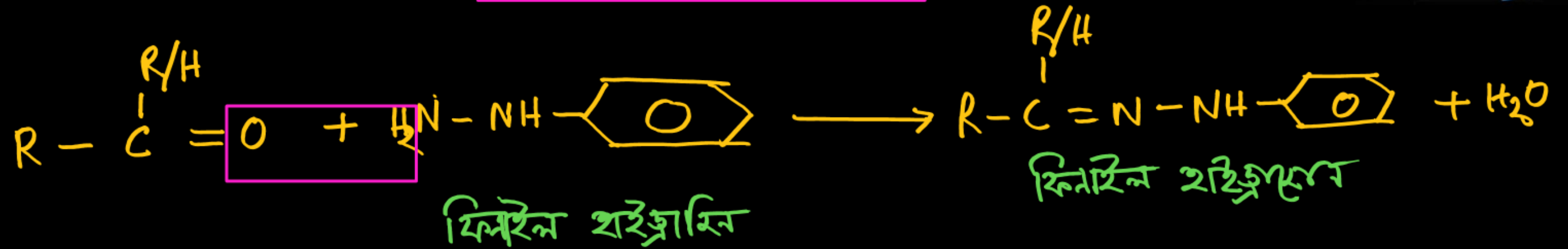
- 1) Nu^- সংযোজন
- 2) ঘনীভবন (DNP Test)
- 3) জারণ (টলেন, ফেলিং) হ্যাঁ
- 4) বিজারণ (ক্লিন্সম্যান)
- 5) অ্যাক্সের সাথে বিক্রিয়া (অ্যান্ডল / হ্যান্সল)
- 6) হ্যান্সলিফেশন
- 7) পলিমারকরণ

1) Nu অণু যোগন:



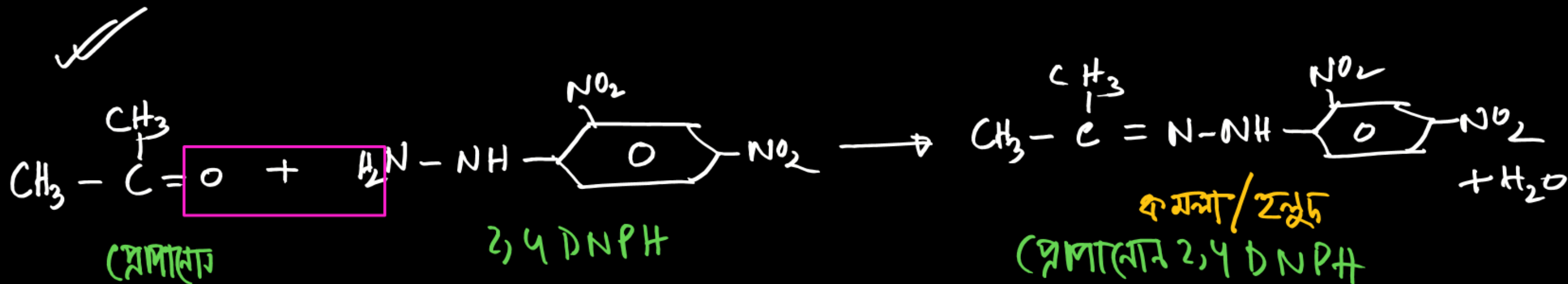
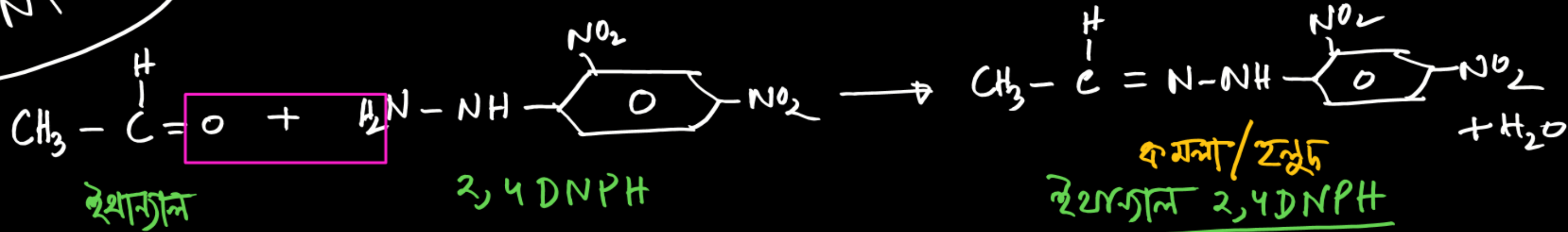
2) ঘনীভবন

NH₂



DNP Test

DNP Test



অ্যালডিহাইড ও কিটোন

$\text{—}\overset{\text{O}}{\parallel}\text{C—}$ এর সমাপ্তকারী পদার্থ

অ্যালডিহাইড + DNPH $\longrightarrow$ কমলা/হালুদী ✓

কিটোন + DNPH $\longrightarrow$ কমলা/হালুদী ✓

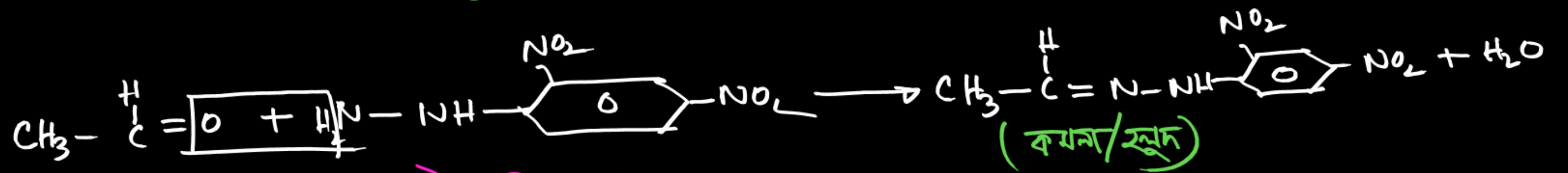
DNP Test

+
অ্যালডিহাইড/কিটোন

* ইথান্যাল ও ইথানয়িক এস দুটি পার্থক্যের
বিশিষ্ট লিখ।

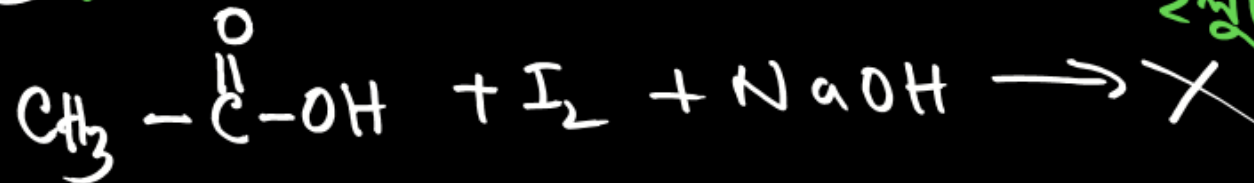
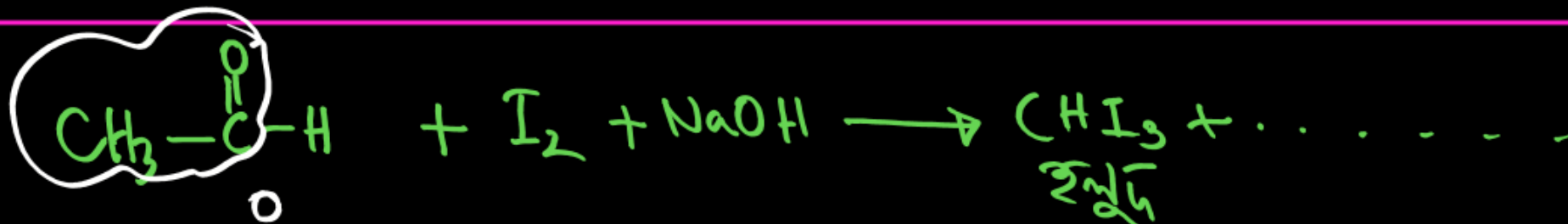
①

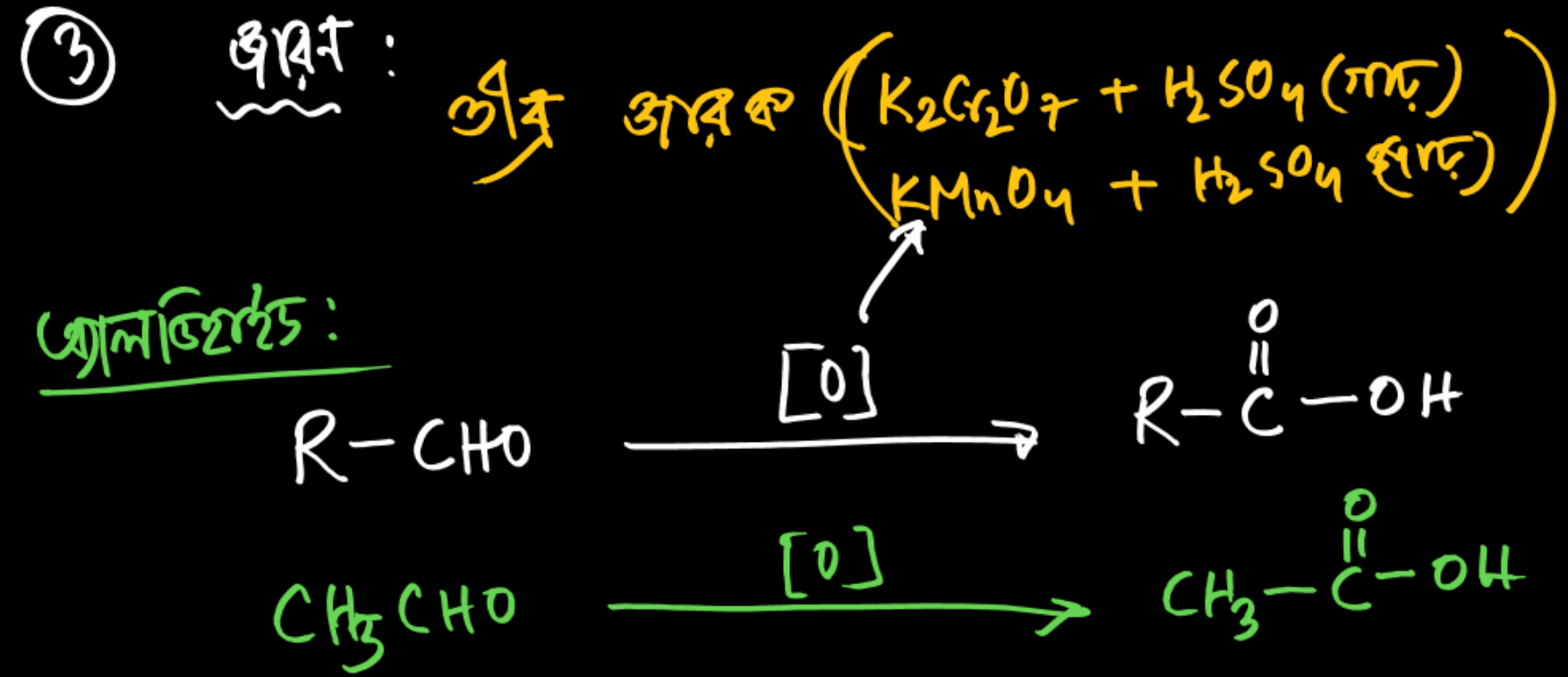
ইথান্যাল + DNPH $\longrightarrow$ ✓



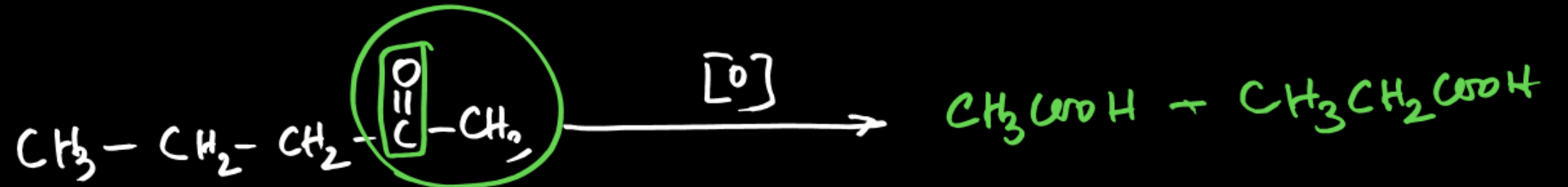
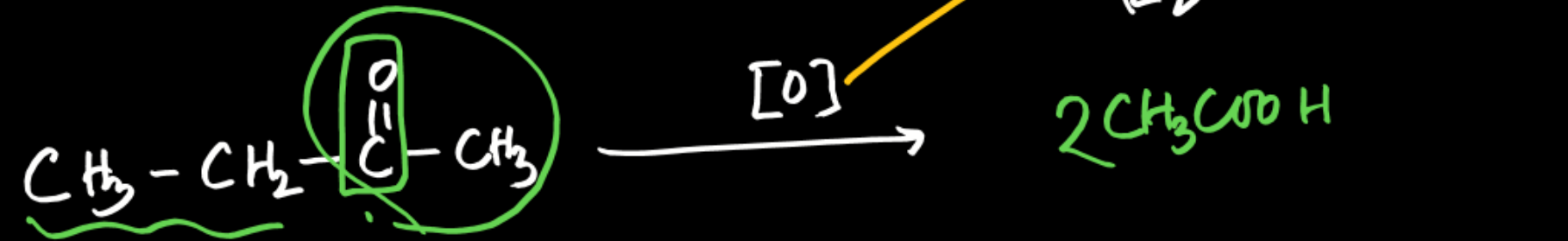
ইথানয়িক এসিড + DNPH $\longrightarrow$ X

②

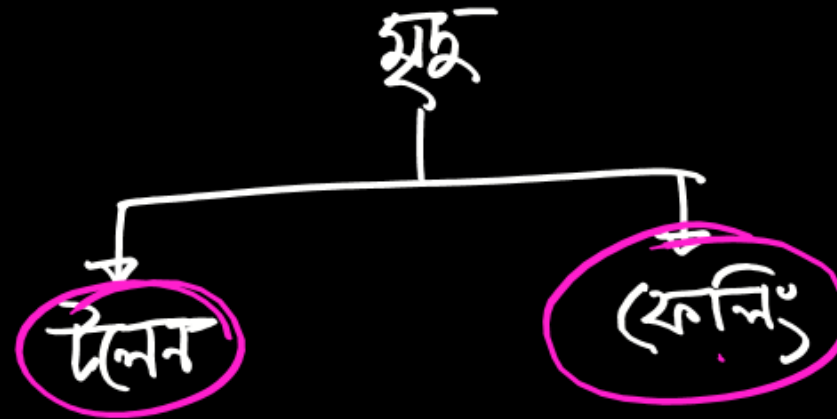




কিটোন:



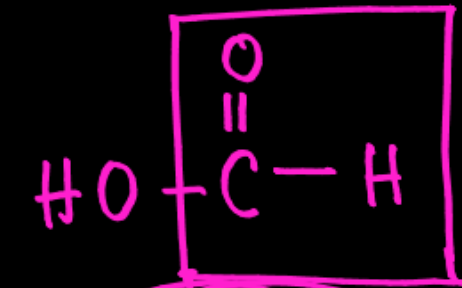
কিটোনগুলি
একই জোড়
স্থানে
 CO_2
জন্ম নেয়।



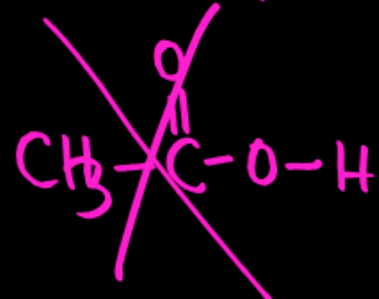
✓ অ্যালডিহাইড + শ্রুত ভারক → টেব এমিড

কিটোন + শ্রুত → ✗

অ্যালডিহাইড ও কিটোন পার্থক্যকারী বিক্রিয়া
টলিন, ফেনিল

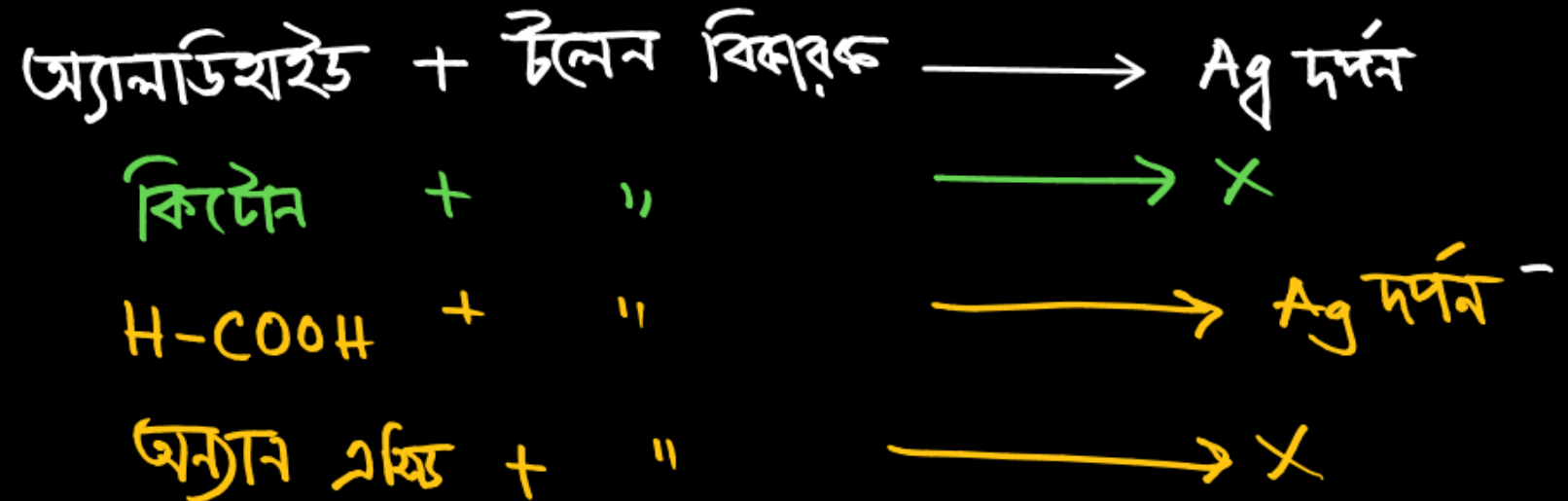
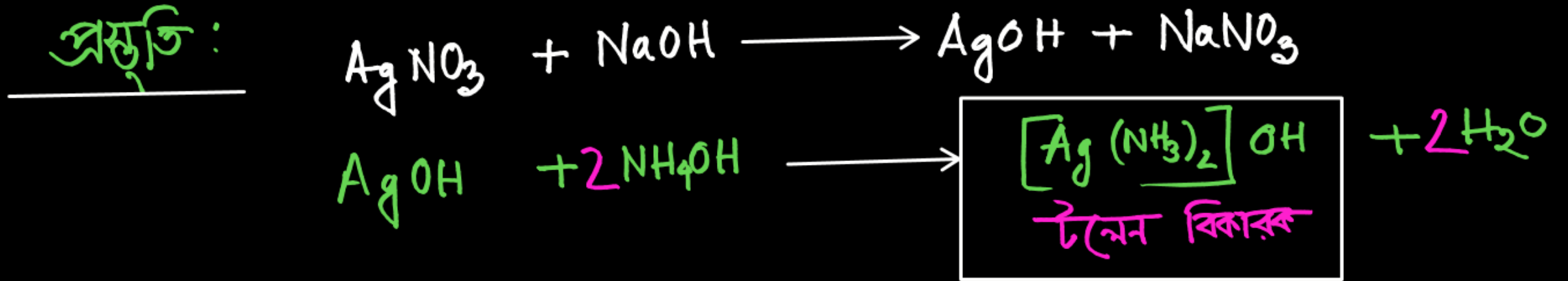


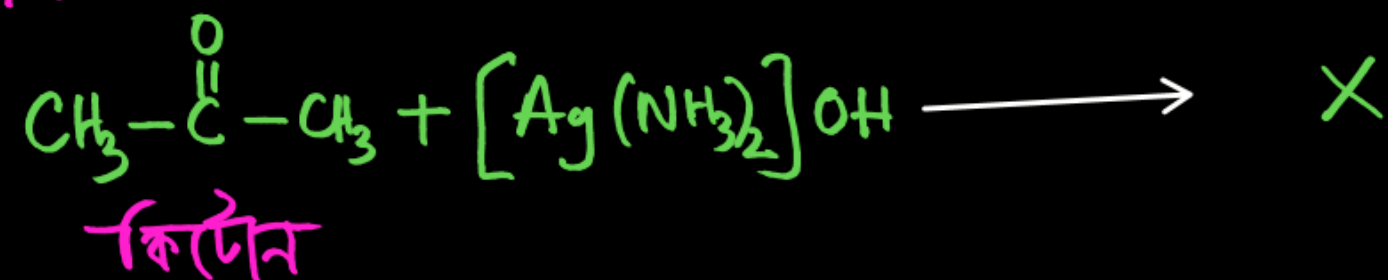
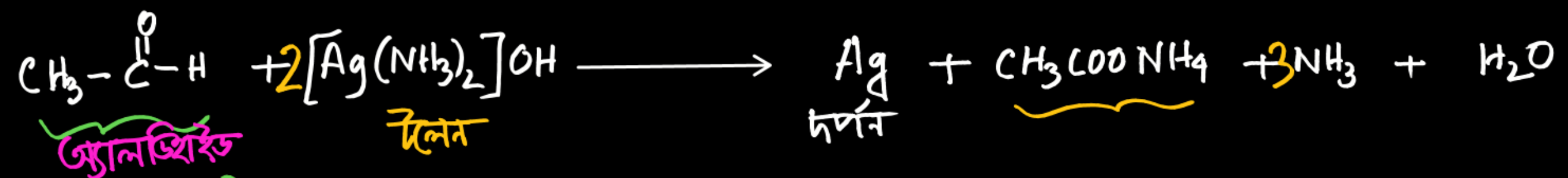
HCOOH
মিথানপিক এসিড



→ টলিন দ্বারা ব্যক্তি
শ্রুত

টলেন বিকারক : $[Ag(NH_3)_2]OH$





ফেনিল: দ্রব:

কপার সালফেট
(5%)

NaK টারট্রেট এর অম্লীয়তা মিশ্রণ
(5% NaOH মিশ্রিত)
(গোচিনি লবণ)

COOH

CH-OH

CH-OH

COOH

টারটারিক অম্ল

COO Na

CH-OH

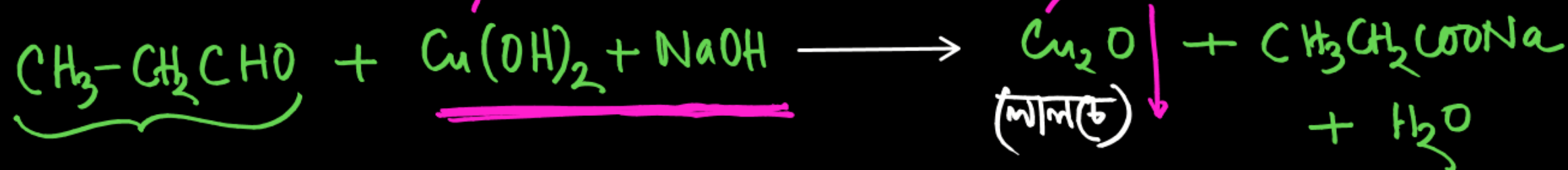
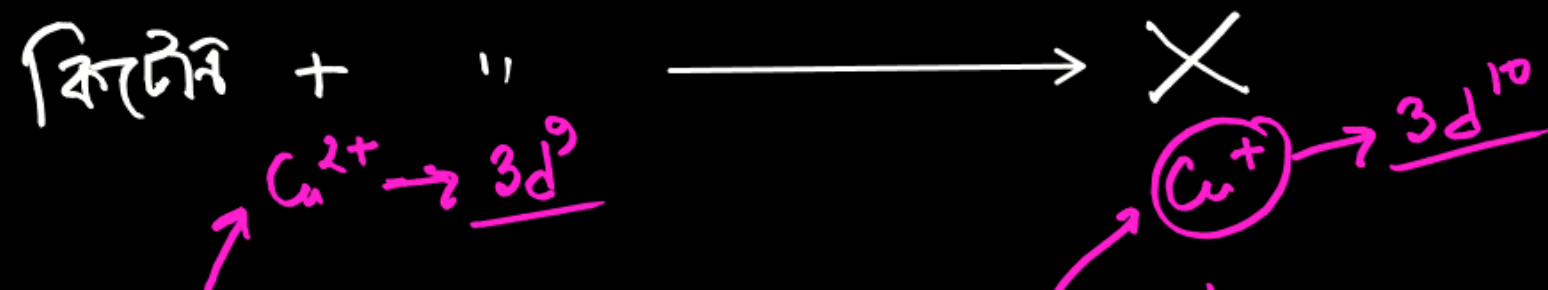
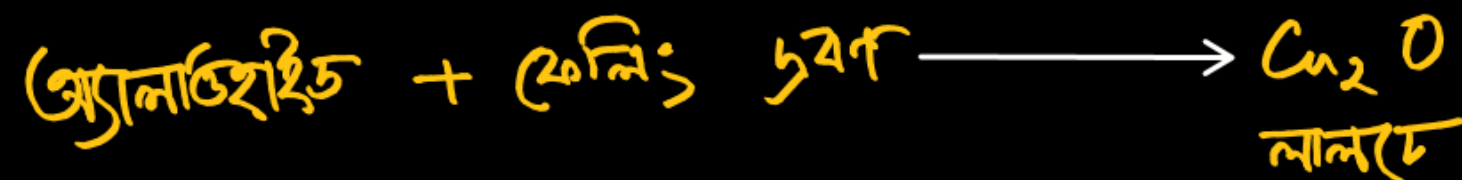
CH-OH

COO K

NaK টারট্রেট
($\text{NaKC}_4\text{H}_4\text{O}_6$)

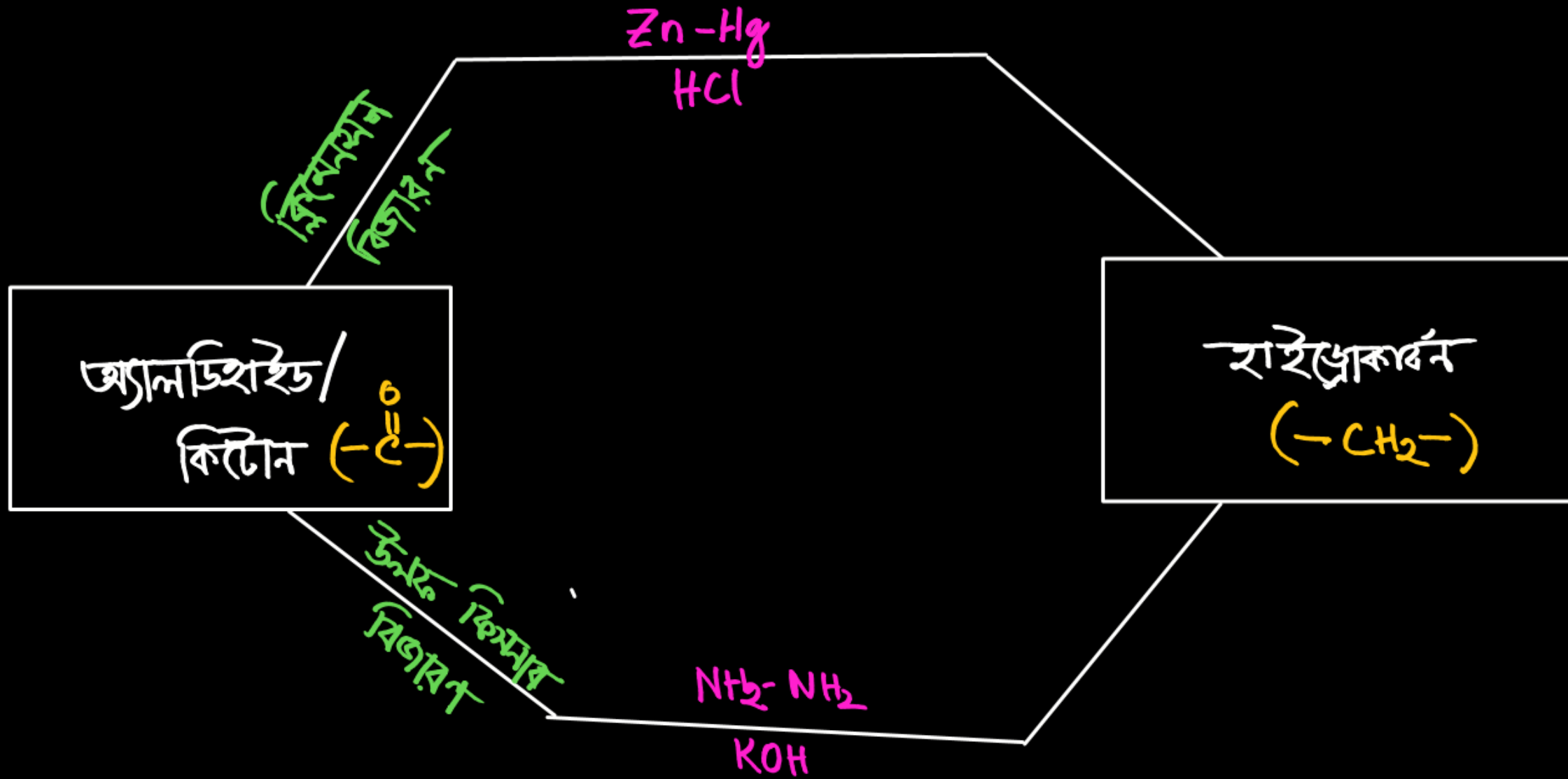


গাঢ় নীল বর্ণের
ফেনিল দ্রব

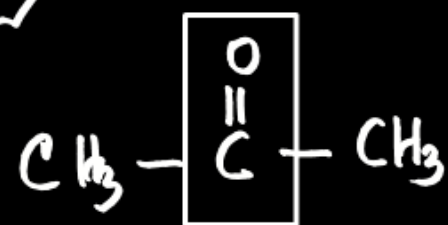


$\text{CH}_3 - \text{CH}_2 - \overset{\text{O}}{\parallel} \text{C} - \text{H} \xrightarrow[\text{LiAlH}_4 / \text{NaBH}_4 / \text{Na-Hg} \xrightarrow{\text{H}_2\text{O}}]{[\text{H}]}$
 $\text{CH}_3 - \text{CH}_2 - \text{CH}_2\text{OH}$
 $\text{CH}_3 - \overset{\text{O}}{\parallel} \text{C} - \text{CH}_3 \xrightarrow[\text{LiAlH}_4 / \text{NaBH}_4 / \text{Na-Hg} \xrightarrow{\text{H}_2\text{O}}]{[\text{H}]}$
 $\text{CH}_3 - \overset{\text{OH}}{\underset{|}{\text{CH}}} - \text{CH}_3$

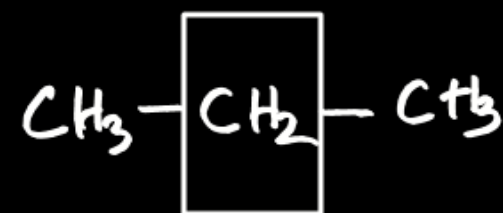
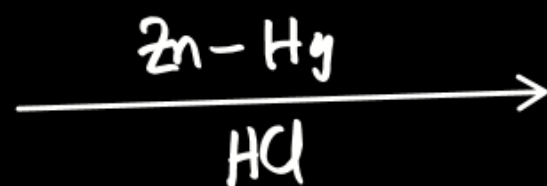
ঐক্য বিজারক দ্বারা বিজারন



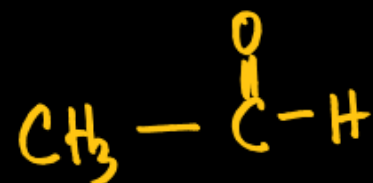
সিঙ্গলিনাম



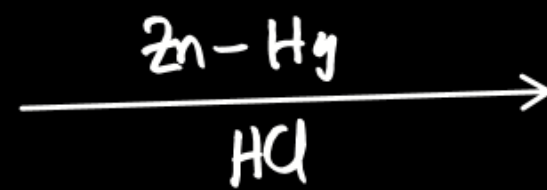
প্রপানোন



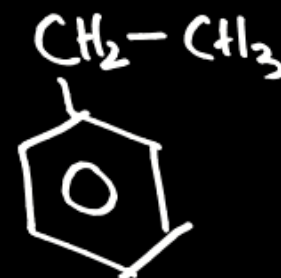
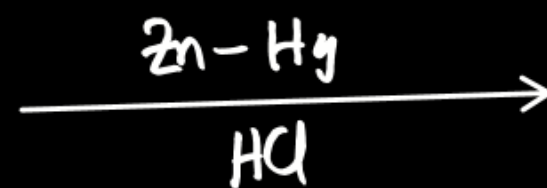
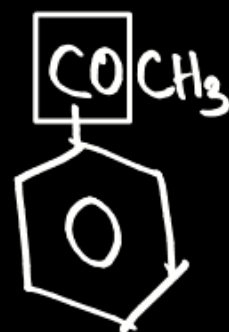
(প্রোপেন)



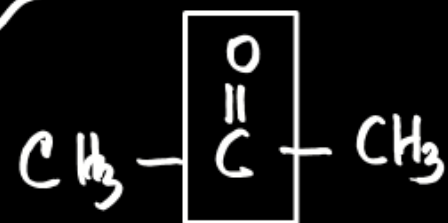
ইথানোন



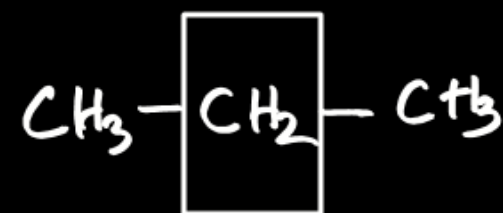
ইথেন



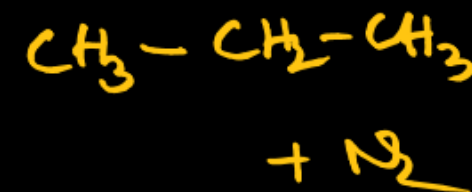
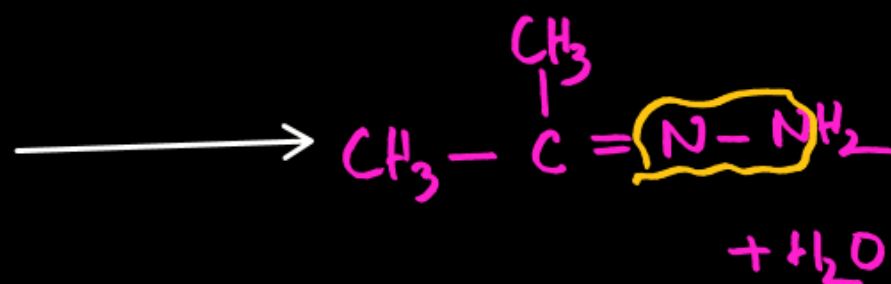
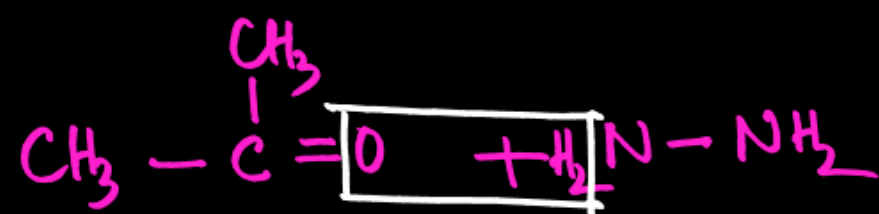
ବ୍ରୋ-ବିକେଟ



କ୍ରମାନ୍ତର



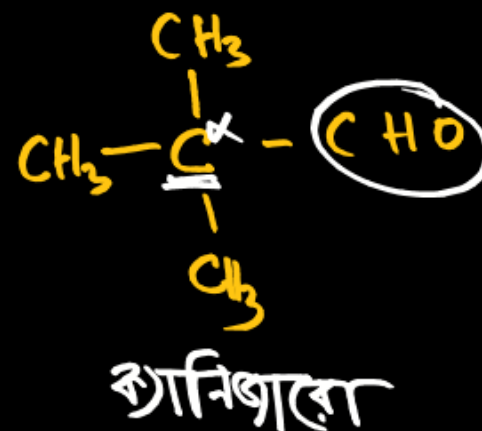
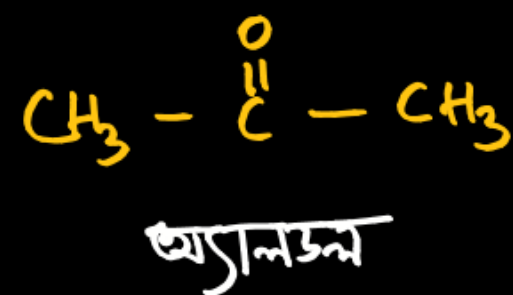
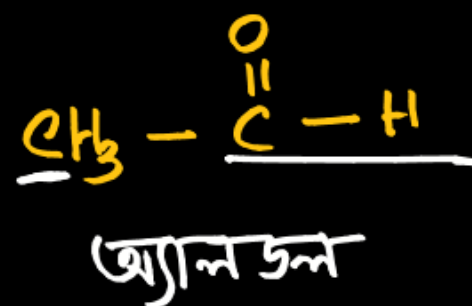
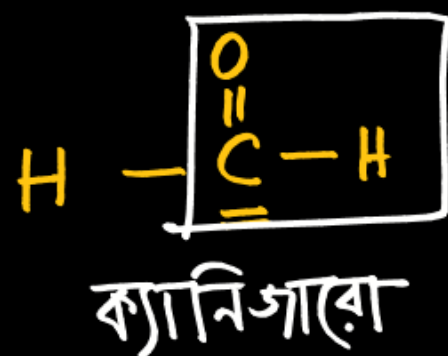
(ପ୍ରାପ୍ତ)



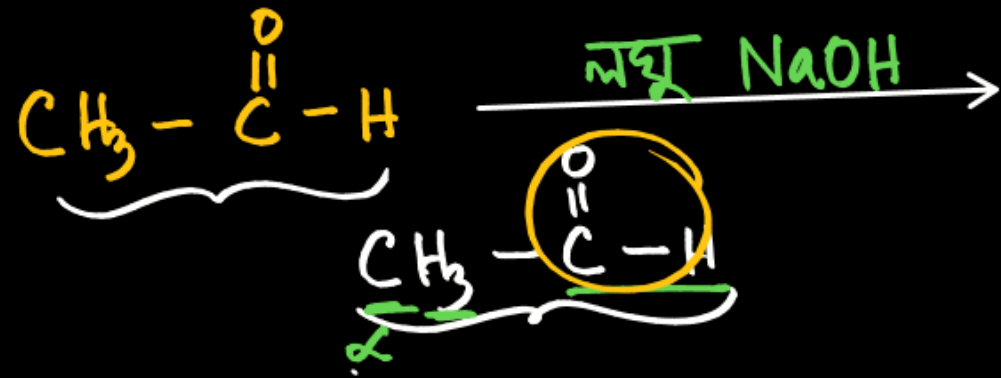
(5) ଆବେଶ ଆଦେଶ ବିଶ୍ଳେଷଣ

ଆଲଡିଲ (α -H ଥିବା + ଲଡ଼ ଆବେଶ)

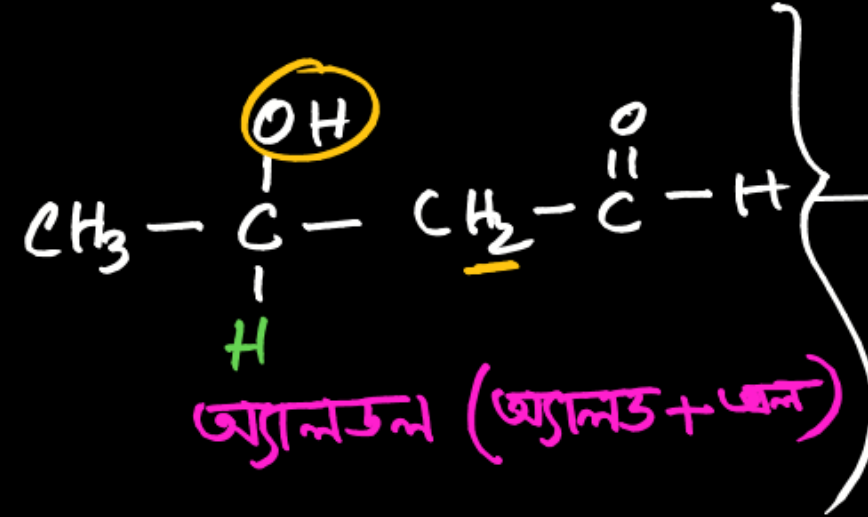
କ୍ୟାଲିକ୍ସାନ୍ (α -H ବିହୀନ + ଗାଢ଼ ଆବେଶ)



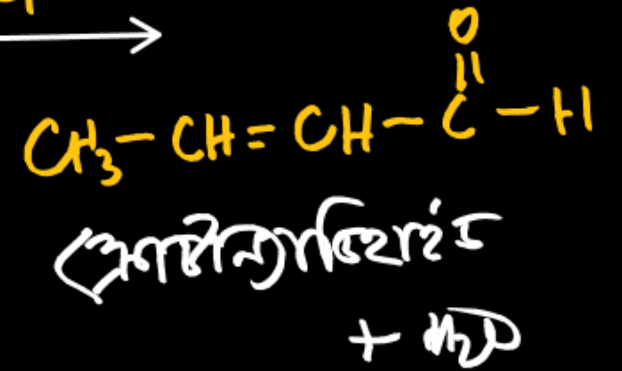
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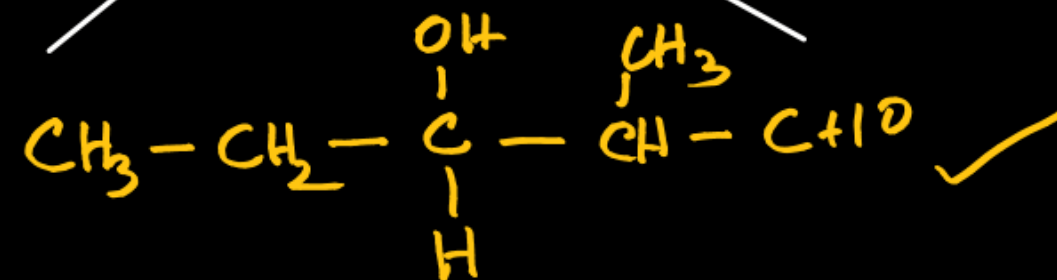
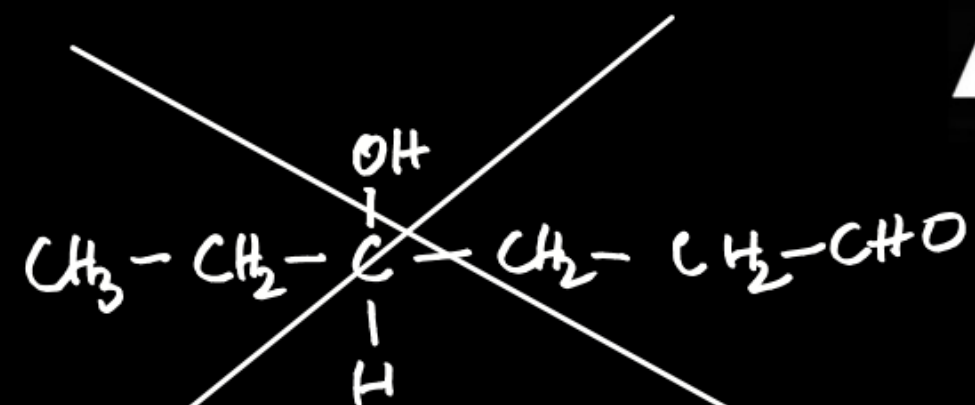


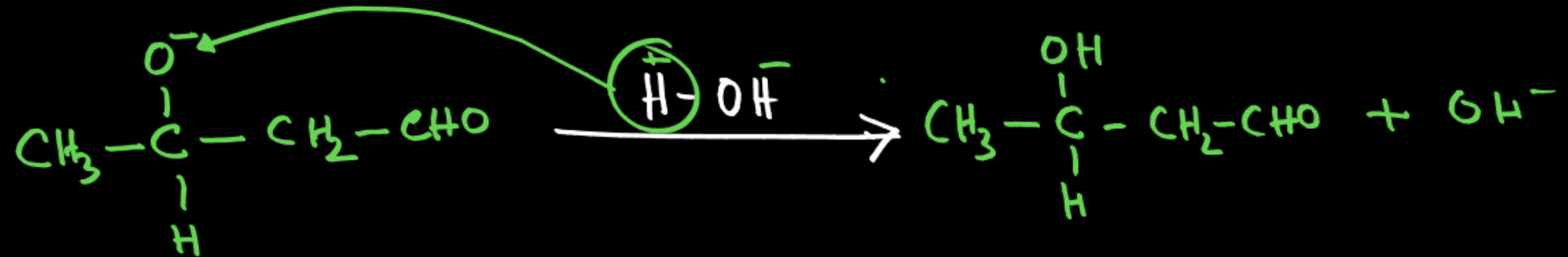
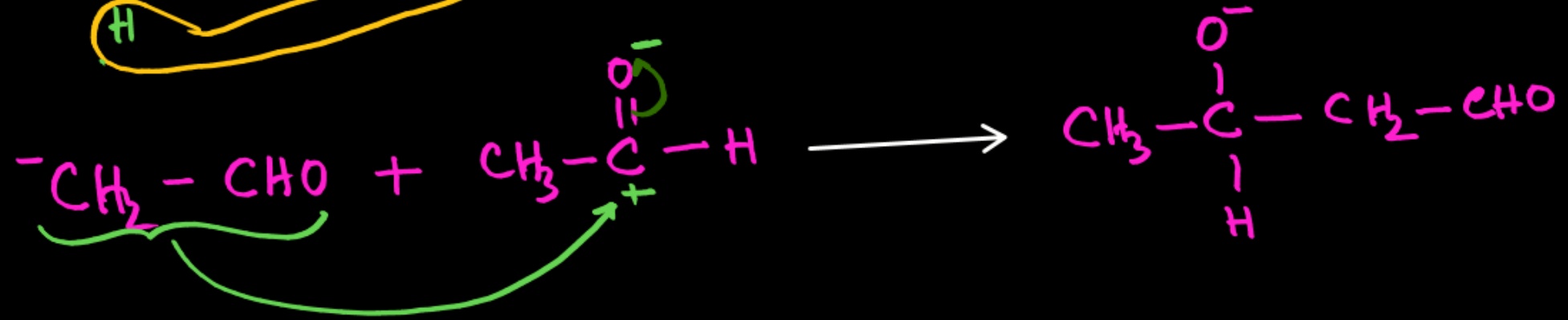
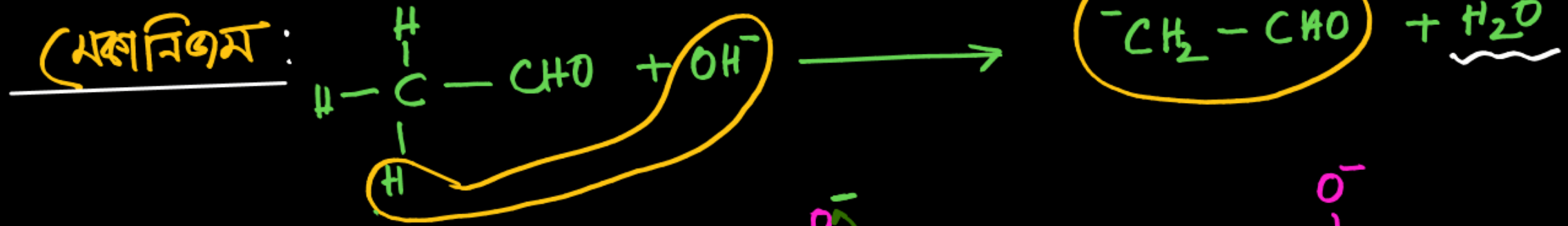
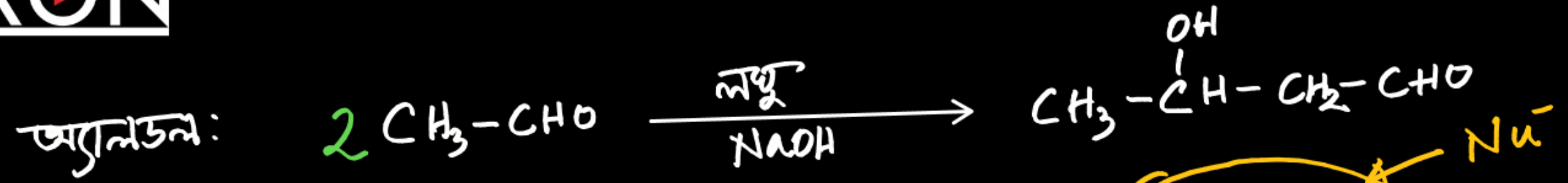
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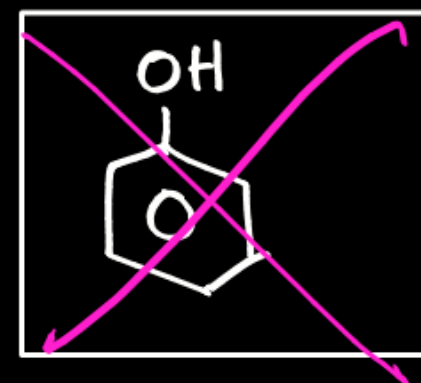
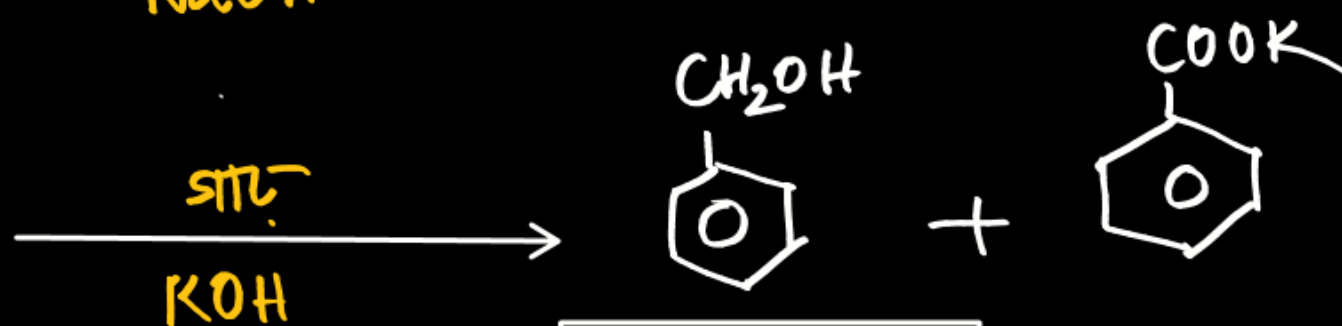
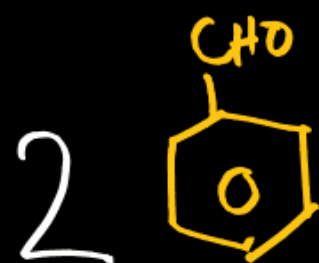
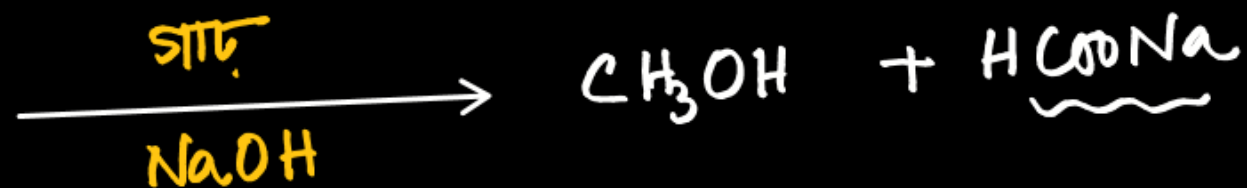
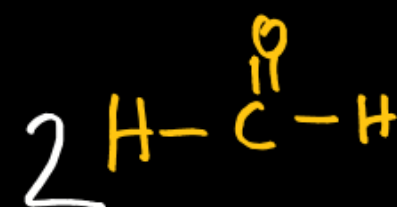
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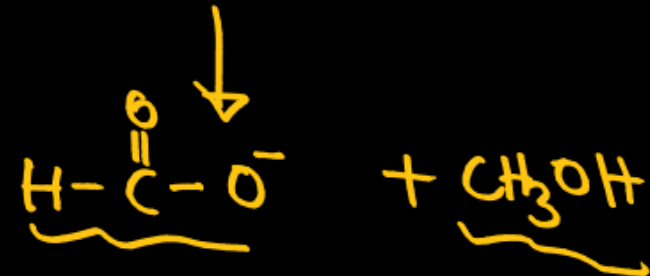
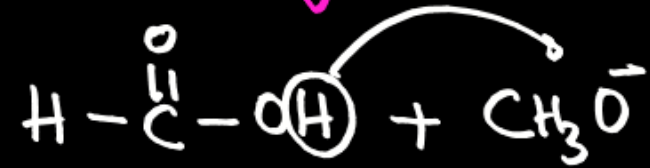
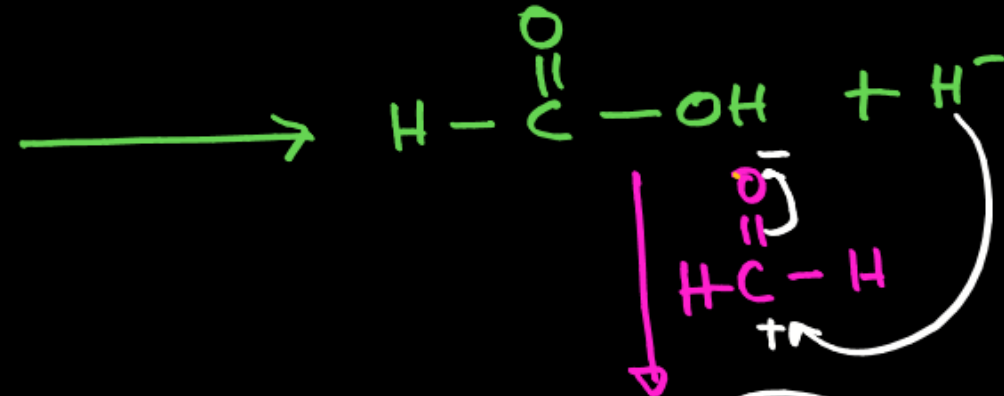
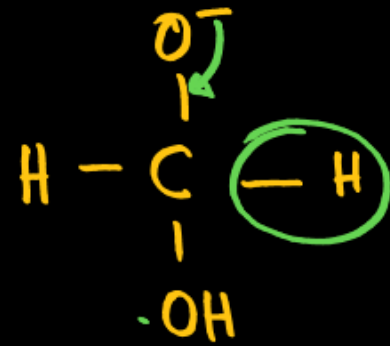
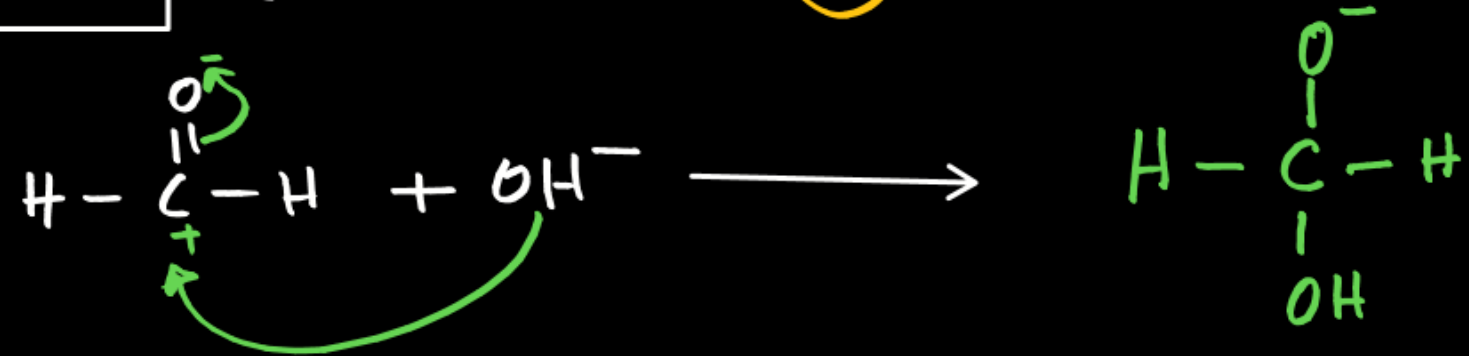


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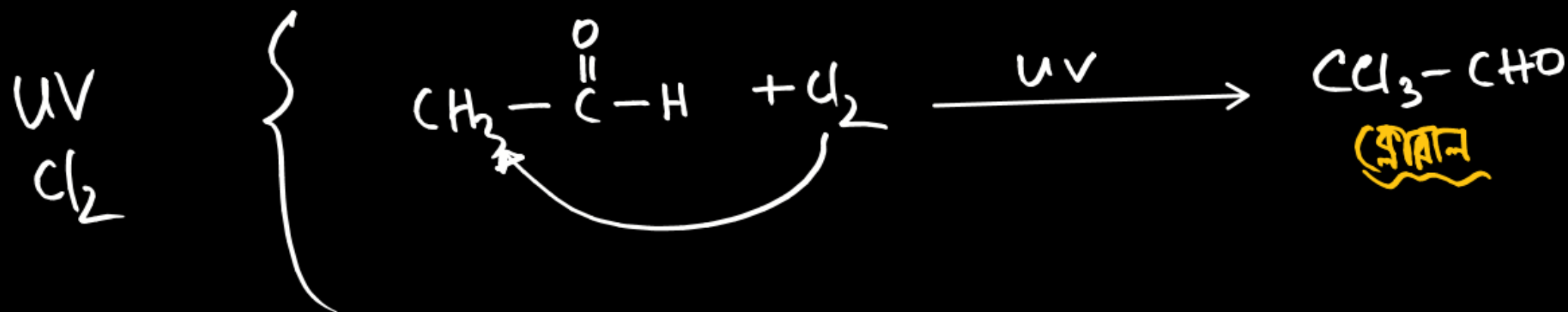
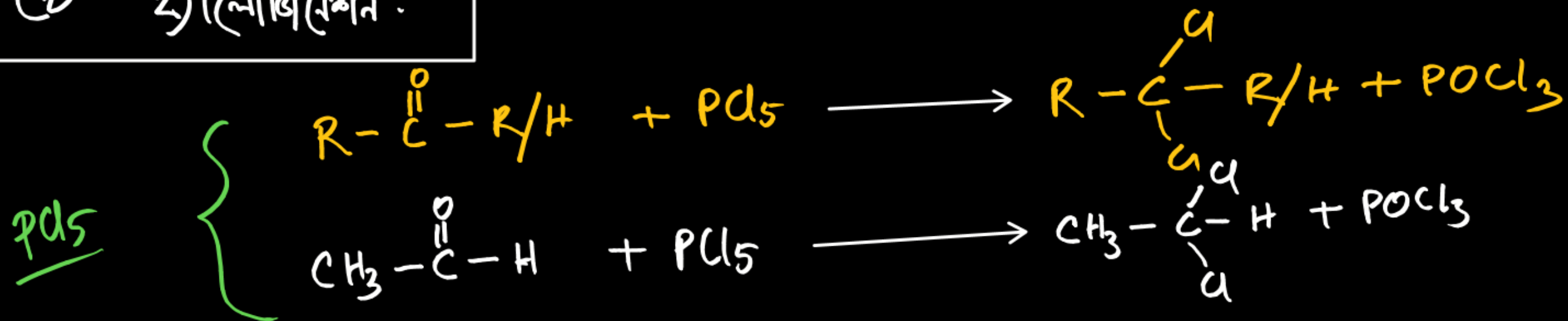




(মেকানিজম):



(3) 2) (ଲବିଷ୍ଟ):



Most Important

1. Nu অণুসমন বিক্রিয়াকারী অণু
2. DNP Test
3. টেন্ড 3 ফেনিঃ
4. ক্লিয়ারমেন 3 টেন্ড ফিনাৰ-
5. অ্যান্‌লিন 3 ক্যানিফাৰ
6. হ্যালোফেন